

Statistical Analysis Report

Aspect-Based Sentiment Analysis on Laptop Reviews (Flipkart) — Week 06

Overview

This report presents the results of hypothesis testing conducted on the cleaned and enriched laptop reviews dataset (16,991 rows, 365 products, 20 brands). Three statistical questions were formulated based on patterns observed during exploratory data analysis (Week 05), and appropriate non-parametric tests were applied given the heavily skewed distribution of star ratings (58% five-star). Each question follows the same structure: business question, hypotheses, test selection and reasoning, results, and business interpretation.

Question 1: Does average rating differ across brands?

Hypotheses

H0: Average rating is the same across all brands.

H1: Average rating differs across at least one brand.

Test Used

Kruskal-Wallis test. Chosen over a standard one-way ANOVA because ratings are heavily skewed rather than normally distributed, violating ANOVA's core assumption.

Results

Metric	All Brands	Filtered (min. 30 reviews)
H-statistic	309.18	308.60
p-value	2.62e-54	4.21e-56

Three brands (Avita, GIGABYTE, Thomson) had fewer than 30 reviews and were excluded in the filtered test to confirm the result was not driven by unstable, small-sample averages. The result held: p remained far below 0.001 in both cases.

Interpretation

There is strong statistical evidence that average rating differs across laptop brands. Established, high-volume brands such as Apple, Samsung, and Primebook cluster near the top (mean ratings above 4.6), while brands such as ZEBRONICS, Ultimus, and walker sit noticeably lower (mean ratings between 3.6 and 3.9), despite having reasonable sample sizes (99 to 116 reviews). This suggests customer satisfaction genuinely varies by brand rather than being uniform noise across the dataset. This analysis does not explain *why* ratings differ by brand — that requires examining what customers are actually saying, which is the purpose of the aspect-based sentiment analysis planned for later phases of this project.

Question 2: Is popularity tier associated with rating?

Hypotheses

H0: Rating is independent of popularity tier.

H1: Rating is associated with popularity tier.

Test Used

Chi-square test of independence. Both popularity tier (Low / Medium / High) and rating (1–5) are categorical variables, making this the appropriate test for checking association between them.

Results

Chi-square statistic	Degrees of Freedom	p-value
128.53	8	5.69e-24

Interpretation

There is strong statistical evidence that rating is associated with popularity tier. High-popularity products show a heavier concentration of 5-star ratings relative to their total volume compared to Low and Medium tiers, which show relatively more 1, 2, and 3-star ratings as a share of their total reviews. This lines up with the Week 05 EDA finding that low-popularity products showed more volatile ratings while high-popularity products stabilized around 4.2 to 4.4. A likely explanation is that products which build up a large number of ratings tend to be ones that have already proven themselves to customers, so sustained popularity and higher satisfaction reinforce each other. This is an association, not a causal claim — popularity does not necessarily cause higher ratings, and higher ratings do not necessarily cause popularity. Both may be driven by underlying product quality.

Question 3: Is there a relationship between review length and rating?

Hypotheses

H0: There is no correlation between review length and rating.

H1: There is a correlation between review length and rating.

Test Used

Spearman correlation. Chosen over Pearson correlation because rating is an ordinal, heavily skewed variable rather than continuous and normally distributed; Spearman measures monotonic relationships without assuming normality.

Results

Spearman's rho	p-value
-0.112	2.66e-48

Interpretation

There is a statistically significant negative correlation between review length and rating ($p < 0.001$), so H0 is technically rejected. However, the correlation coefficient itself (-0.112) is very weak and close to zero, meaning review length explains almost none of the variation in rating. This is a clear example of statistical significance versus business significance: with 16,991 reviews, even a very small, practically meaningless relationship can produce a very small p-value, because p-values are sensitive to sample size, not just effect size. The correct business takeaway is not “longer reviews mean lower ratings,” but rather “review length is not a meaningful predictor of rating on its own.” This is consistent with the Week 05 EDA finding of near-zero correlation between review length and star rating, now confirmed with a formal significance test rather than a visual impression alone.

Conclusion

Three hypothesis tests were applied to validate patterns observed during Week 05 EDA. Average rating differs significantly across brands, even after excluding brands with fewer than 30 reviews. Rating is significantly associated with popularity tier, consistent with the EDA finding that low-popularity products show more volatile ratings. Review length shows a statistically significant but practically negligible relationship with rating, illustrating the distinction between statistical and business significance.

Together, these results confirm that brand and product popularity are meaningfully linked to customer satisfaction, while review length is not. These findings support moving into the next phase of the project, Aspect Extraction, where the focus shifts from *whether* ratings differ to *understanding why*, by identifying which specific product aspects (battery, display, build quality, etc.) drive positive or negative sentiment.